

Exploring Simulated Web Agents vs Real User Journeys

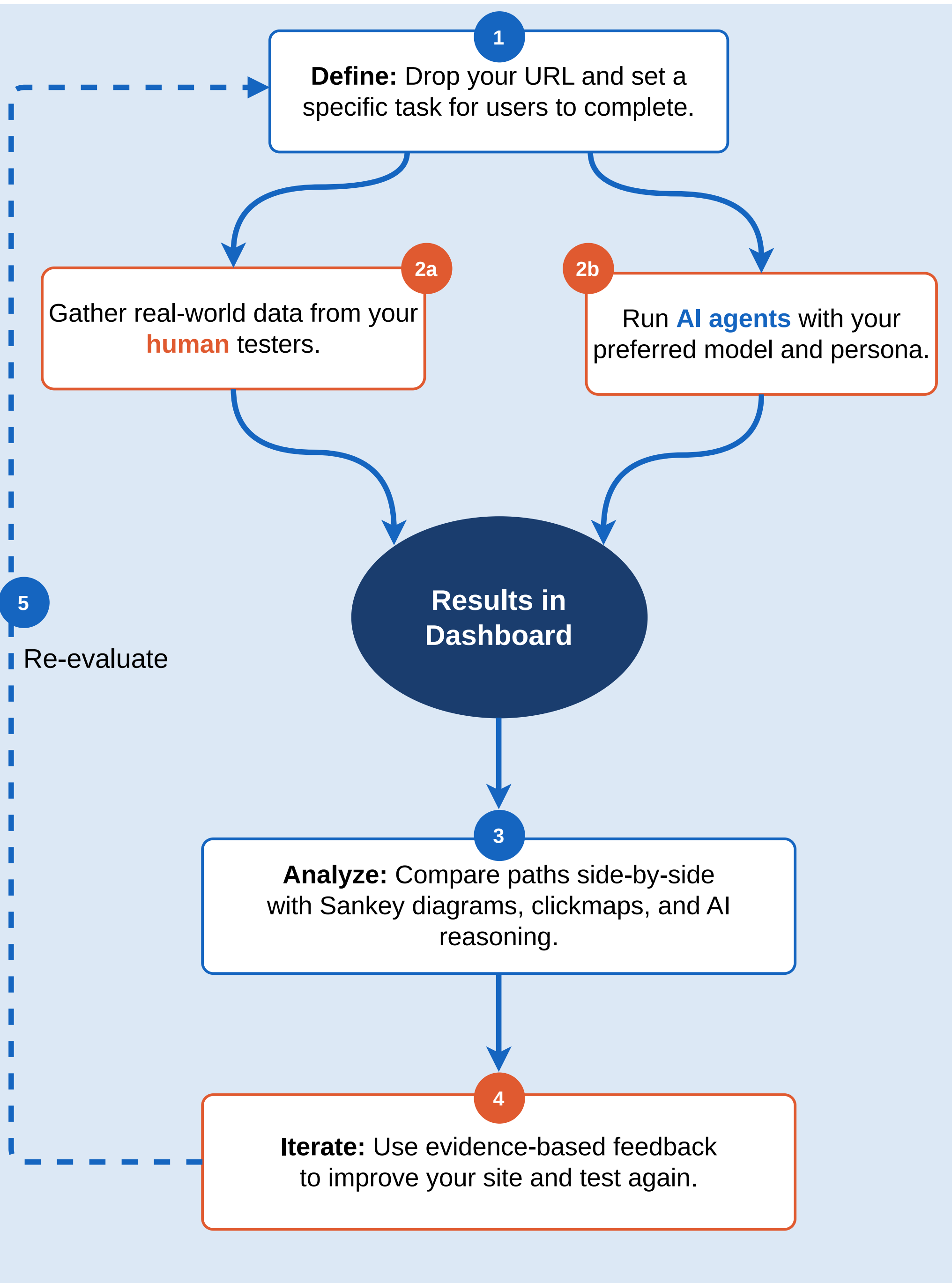
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1 Introduction

We propose a human-centered UX evaluation framework leveraging webagents and human journey data. By aligning the webagents to human navigation behaviour, we enable scalable and faithful UX testing without iterative manual data collection.

- **Human vs Agent trajectory:** Comparison reveals divergence and alignment invisible to automated testing.
- **Scalable re-evaluation:** Re-usable human data steers agents and enables iterative design testing at scale.

2 Workflow



3 Users & Data

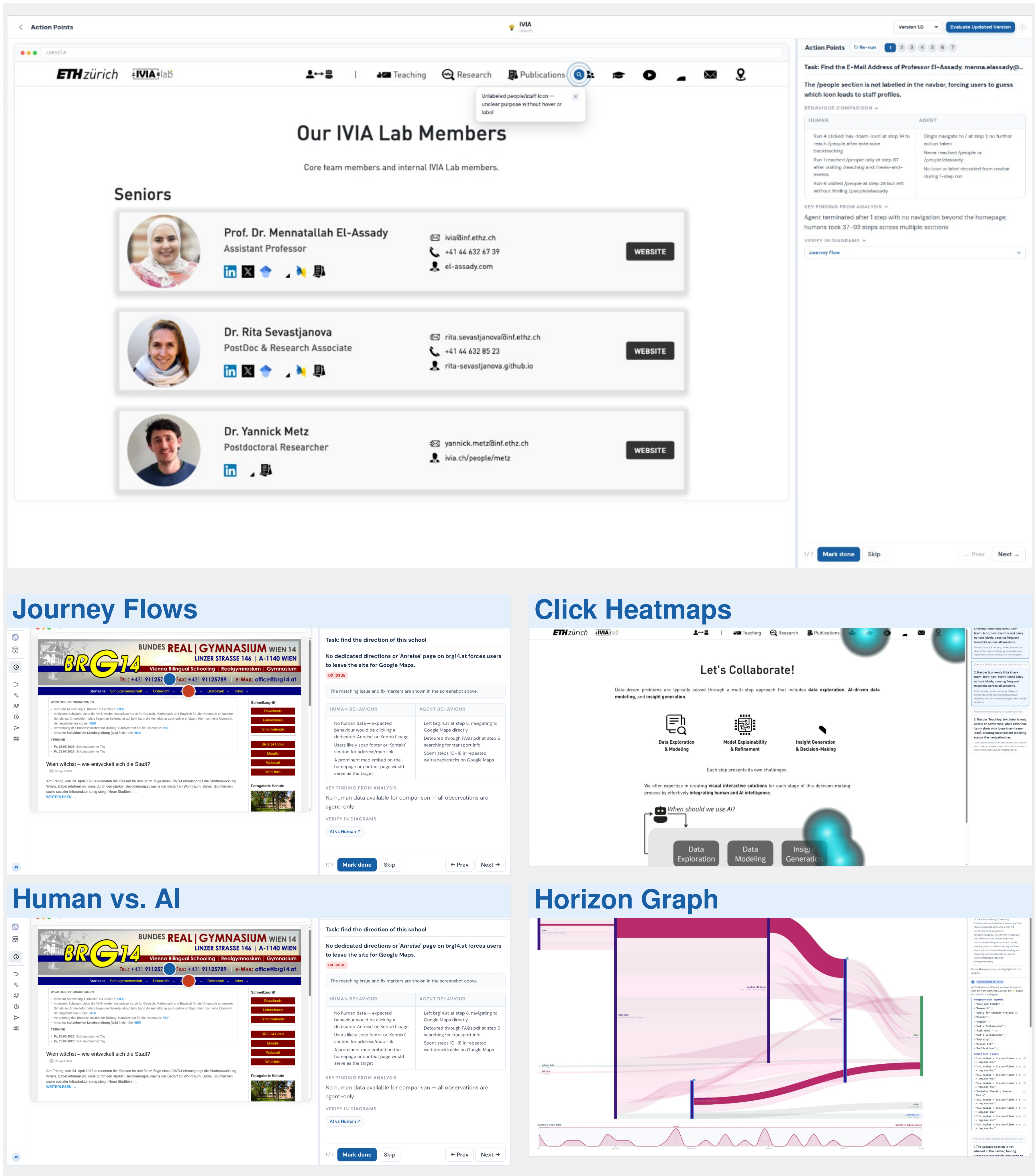
Users

- **UX designers & Web developers:** The designer can use the tool to evaluate their design and human interaction patterns simulated by the aligned webagents.

Data

- **Human journeys:** We generate a testing link to be sent to the human testers, who explore the website accomplishing predefined tasks. Their website interaction is captured via a reverse-proxy tracking script.
- **Agent journeys:** Browser-Use agents with personalized system prompts solve the tasks and their navigation logs are stored.
- **Data-Types:** Clicks, Scrolls, Screenshots, Timestamps, URLs, DOM events

4 User Interface



5 Agent Steering & Evaluation

Dashboard Analysis

- **User Interface:** Ranked usability issues anchored to screenshots and linked to visualisations.
- **Aggregate Journeys:** Flow level path divergence.
- **Heat Map & Horizon:** Attentional differences and interaction intensity over time.
- **Human vs. AI:** Statistical comparison of human and agent behaviour.
- **Flow & Sankey:** Navigation patterns and click-level path divergence.

Agent Steering & HITL Re-evaluation

- **Prompt-injection steering:** Human trajectories injected as context to bias agents toward human-like navigation without fine-tuning.
- **Re-evaluation:** Steered re-runs quantify alignment across versions without new recruitment.

6 Conclusion

Our framework demonstrates that webagents can be used to simulate human navigation behaviour. This enables scalable UX evaluation and iterative design testing without the need for repeated human data collection, which can be costly and time-consuming. While unaligned agents diverge, we manage with simple prompt-injection to steer agents toward realistic navigation patterns customized for each specific website. LLM-powered agents steered by human trajectory priors serve as scalable proxies for UX evaluation. Prompt-injection alignment preserves behavioural fidelity while reducing manual testing cost. Future work targets VLM fine-tuning and automated action-point propagation.

